Flying Straight

Pupil sheet

Name: _____ **Date:** _____

Safety: I hold the box in both hands, over the table. The cable stays slack. I never throw the box.

1. What I see on the screen

On the screen you see two numbers — **Roll** (tilt to one side) and **Pitch** (tilt forwards and back) — and a **circle** that turns **green** when both are below 5°.

The level. The box is **set down** in the middle of the surface, with the same edge towards me. **Nobody touches the table**, and **the cable lies slack**, because a taut cable pulls the box crooked.

The challenge: I start with the box on the desk, lift it **in both hands**, elbows on the table, and keep the circle green for **ten seconds**. If I leave the green, **the timer starts again from zero**.

2. What I do on my own

First I predict. I do not change my prediction after I measure.
I put them in order, from the most level to the most tilted — *what I think*:

1. _____ 2. _____ 3. _____ 4. _____

I think so because _____

_____

Now I measure. I put the box **in the middle** of the surface, with the same edge towards me every time, and take my hands off it.

Surface	Roll (°)	Pitch (°)	Does it go green?
My desk			yes / no
The window sill			yes / no
The floor			yes / no
With a book underneath			yes / no

The real order, after measuring:
1. _____ 2. _____ 3. _____ 4. _____

The reversal test. On the most tilted surface I press **Reading 1**, turn the box round on the spot, the other end towards me, and press **Reading 2 (turned round)**. The number: ☐ changes sign (the surface is tilted) ☐ stays the same (it is the box)

The challenge. The number of attempts is not an embarrassment — what counts is what I changed between them.

Attempt	I held it (s)	What I changed before the next one
1		
2		
3		
4		

3. The conclusion I write

I predicted that _____

_____

The data show that _____

_____

A pilot who can see nothing through the window relies on _____

_____

Exit ticket

What did you change between attempts?

_____